Two ways AI “breaks” Navier-Stokes

“AI broke Navier-Stokes”

AI as prover

finds a proof about the equations

AI as stand-in

replaces the expensive solve

AIMLC Meetup · Capital One Flatiron · Nov 3 · Platon Karpov, hosted by Vladimir Eremin

Four accepted outcomes. Which one?

NO EXTERNAL FORCE

A, B prove smooth solutions exist for all time

the version many mathematicians call the deeper question

SMOOTH EXTERNAL FORCE ALLOWED

C, D show a smooth flow can blow up

OpenAI's announced result maps here (per DataCamp)

> Domain of OpenAI's result (whole space vs periodic box): not stated in the coverage read.

Source: Clay Millennium Prize statement (Fefferman); DataCamp, Sept 2026

What the checker checks

Clay problem, in English

GATE 1

Human judgment: is the formalization faithful?

UNVERIFIED · No independent verification reported (as of Sept coverage)

Formal statement

GATE 2

Lean: does every step follow? (OpenAI says yes)

Proof

Lean certifies the logic, not the question.

Source: Science News (Eyink, Johns Hopkins), Sept 2026; The Neuron, Sept 2026

Prover vs stand-in: mirror-image problems

	PROVER	STAND-IN
HAS A CHECKER?	**Yes:** Lean	**No:** conservation checks, error bars, benchmarks
IS THE QUESTION CLEAR?	**Contested:** forced vs unforced	**Yes:** match the solver
OPEN RISK	formal statement ≠ intended problem	neural operators, TurtLES – not Sapsan loses small-scale energy, unstable on long 3D rollouts
EXAMPLE	OpenAI, forced 3D NS (announced Sept 8)	Sapsan, supernova turbulence (Karpov et al., JOSS 2021)

Sources: NeurIPS 2026 TurtLES poster; arXiv 2601.22541; JOSS 2021; DataCamp 2026

Right, or quietly wrong?

When a learned model replaces a solver, how do you know?

Asking **Platon Karpov** on Nov 3. **Send me yours.**